Database Management Systems
Assignment-2

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Section-A

Ques1 - Fill in the blanks -

1. The Entity Relationship (ER) Diagram visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

2. In the Relational Model, data is organized into tables, with relationships b/w entities defined through foreign key constraints.

3. The Object Oriented Model treats data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions.

4. Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaing the accuracy of information.

Section-B

Ques2 - In a healthcare system, discuss how data abstraction can simplify the interaction b/w healthcare professionals and patient records.

Ans - Data abstraction refers to the process of hiding complex data structures and showing only the essential information to users.

In a healthcare system, data abstraction simplifies the interaction b/w healthcare professionals and patient records in the following ways:

a) Simplified access to information : Doctors and nurses do not need to understand the complex database structure. They can simple view patient details such as name, age, diagnosis and treatment through an easy interface.

b) Different views for different users : Different ~~xx~~ healthcare professionals require different information.

- Doctors see medical history and prescriptions.
- Nurses see medication schedules and patient states.
- Administrative staff see billing and appointment details.

c) Improved Data Security: Sensitive patient data is protected because users only see the information relevant to their role.

d) Efficient Data Management: Healthcare professionals can quickly retrieve patient data without dealing with technical database operations.

Ques5 - Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Ans - Benefits:

a) Ease of use: Healthcare professionals can easily access patient information without needing technical knowledge.

b) Improved security: Only authorized users can access sensitive medical records.

c) Better Data Management: Large amounts of medical data can be handled efficiently.

d) Reduced Errors: Standardized views reduce the chances of incorrect data interpretation.

e) System Flexibility: Changes in database structure do not affect users because abstraction hides internal details.

Challenges

a) System Complexity: Designing abstraction layers requires careful planning and advanced system design.

b) Implementation Cost: Developing and maintaining such systems can be expensive.

c) Performance Issues: Multiple abstraction layers may sometime slow down system performance.

Section-C

Ques - Design a database for a university using ER model.

Ans - A university database can be designed using three main entities.

1. Student Entity

Attributes:

- Student_ID (Primary Key)
- Student_Name
- Age
- Gender
- Email
- Course_ID (Foreign Key)

2. Course Entity

Attributes:

- Course_ID (Primary Key)
- Course_Name
- Credits
- Department
- Professor_ID (Foreign Key)

3. Professor Entity

Attributes:

- Professor_ID (Primary Key)
- Professor_Name
- Department
- Email
- Phone_number

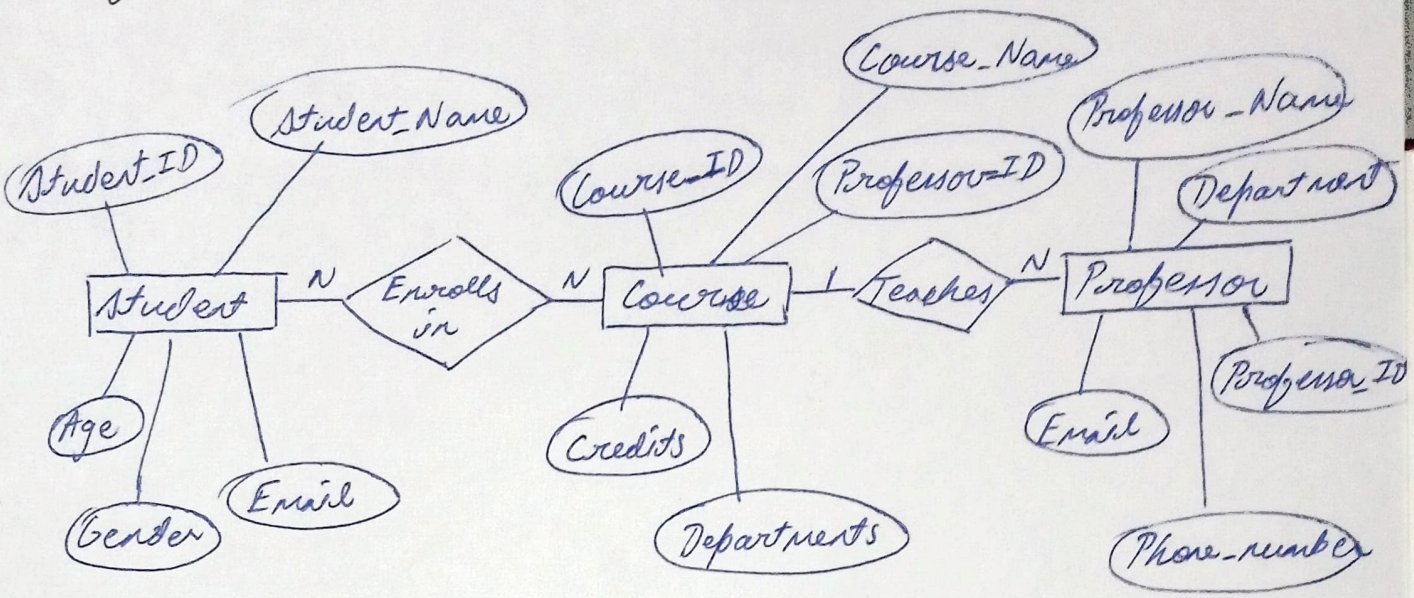
Relationships b/w Entities

1. Student - Enrolls - Course
- One student can enroll in multiple courses.
- One course can have many students.
- This is a many to many relationship.

2. Professor - Teaches - Course
- One professor can teach multiple courses.
- Each course is taught by one professor.
- This is a one to many relationship.

This database structure helps the university manage:
- Student enrollment.
- course details.
- Profession information efficiently and systematically.

ER Diagram:


Student_ID
Student_Name
Student
N
Enrolls in
N
Course
Course_ID
Course_Name
Professor_ID
Teaches
1
N
Professor
Professor_Name
Department
Professor_ID
Age
Gender
Email
Credits
Departments
Email
Phone_number